

Topic 4: Statistics and Probability

Theory & Examples (Sections 4.5 – 4.11)

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1 4.5 Elementary Set Theory

Basic notions

A **set** is a collection of objects (elements), usually denoted by a capital letter.

$\mathbb{R}$ = the set of real numbers

$\mathbb{Q}$ = the set of rational numbers

$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$ (natural numbers)

$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ (integers)

$A = \{1, 2, 3\}$

$B = \{a, b, c, d\}$

$C = \{\text{Chris, Mary, Tom}\}$

- $a \in B$: a belongs to B . $f \notin B$: f does not belong to B .
- The **empty set** contains no elements: $\{\}$ or $\emptyset$.
- **Subsets**: if A contains n elements, there are 2^n subsets.

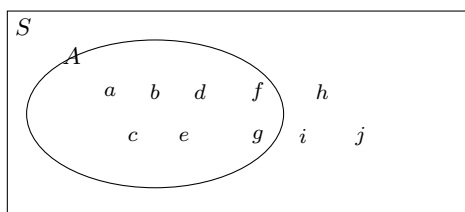
Example. $A = \{1, 2, 3\}$ has 8 subsets: $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$, and $\emptyset$ — since $2^3 = 8$.

- If $A = \{1, 2, 3\}$ and $B = \{1, 2\}$, then B is a **subset** of A : $B \subseteq A$.
- Always: $\emptyset \subseteq A$ (the empty set is a subset of any set) and $A \subseteq A$ (any set is a subset of itself).
- All subsets of A except A itself are called **proper subsets**: $B \subset A$.

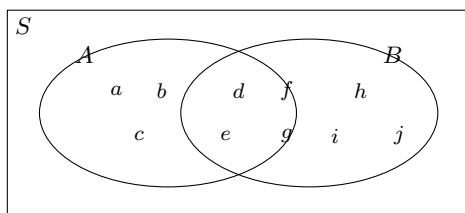
Venn diagrams

We refer to a large set S , called the **universal set**, and consider several subsets of S .

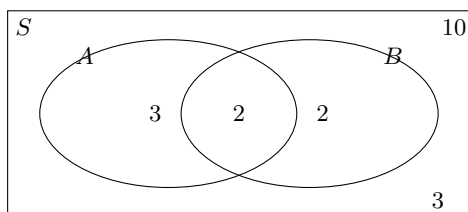
Example. $S = \{a, b, c, d, e, f, g, h, i, j\}$, $A = \{a, b, c, d, e\}$. A helpful way to present this information is by using a Venn diagram:



If we also consider the subset $B = \{d, e, f, g\}$, the Venn diagram becomes:



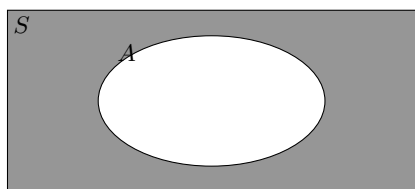
As we usually deal with large universal sets, in a Venn diagram we are not interested so much for the elements themselves but only for the number of elements in each region. In this case the Venn diagram above takes the form:



$n(A)$ = the number of elements of set A . In the example: $n(S) = 10$, $n(A) = 5$, $n(B) = 4$. Notice A consists of two regions (3 and 2 elements), so $n(A) = 3 + 2 = 5$.

The complement of A : A' (not A)

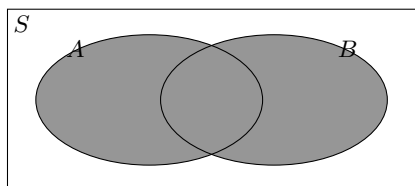
It contains the elements that are not in A .



In our example $A' = \{f, g, h, i, j\}$ [some authors denote the complement of A by $\overline{A}$].

The union $A \cup B$ (read as “ A or B ”)

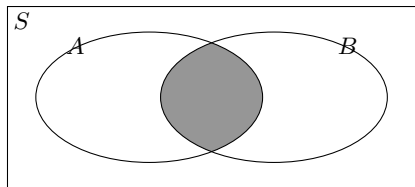
It contains all the elements that belong either to A or to B .



In our example $A \cup B = \{a, b, c, d, e, f, g\}$.

The intersection $A \cap B$ (read as “ A and B ”)

It contains the common elements of A and B .



In our example $A \cap B = \{d, e\}$.

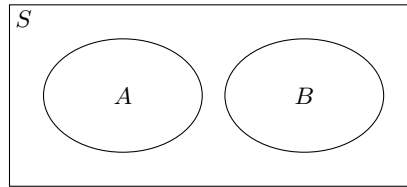
A basic property

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Example. $n(A \cup B) = 7$, $n(A) = 5$, $n(B) = 4$, $n(A \cap B) = 2$. Indeed $A \cup B$ has 7 elements, not $5 + 4 = 9$, because in $n(A) + n(B)$ the common elements are counted twice: $7 = 5 + 4 - 2$.

Mutually exclusive sets

If $A \cap B = \emptyset$, then $n(A \cap B) = 0$:



In this case only:

$$n(A \cup B) = n(A) + n(B)$$

and A, B are said to be **mutually exclusive**.

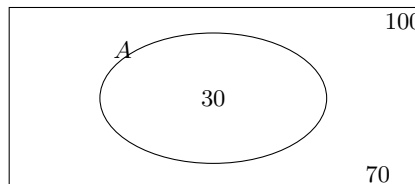
2 4.6 Probability

We start with a universal set S , called the **sample space** in probability theory — it contains all possible outcomes of a game or experiment. Subsets $A, B, \dots$ of S are called **events**.

$n(S) = \text{TOTAL}$. The probability of an event A is defined as:

$$P(A) = n(A) / \text{TOTAL}$$

Example. sample space S has 100 elements, event A has 30 elements $\Rightarrow P(A) = 30/100 = 0.3$. If we choose an element of S at random (all equally likely), the probability it belongs to A is 30%.



$$P(A) = \frac{n(A)}{\text{TOTAL}} = \frac{30}{100} = 0.3$$

Clearly $0 \leq P(A) \leq 1$, and $P(\emptyset) = 0$, $P(S) = 1$.

Complementary events

In the example above, $P(A') = 0.7$. In general:

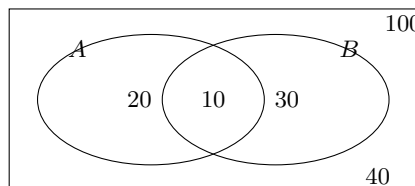
$$P(A') = 1 - P(A)$$

Combined events

Recall $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Dividing all terms by TOTAL:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

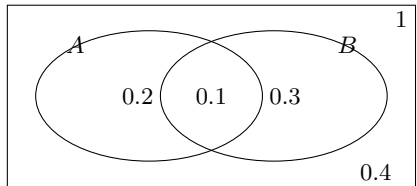
Example. consider:



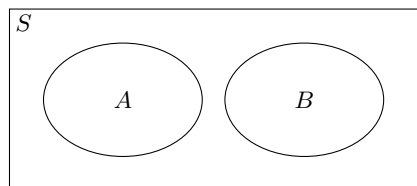
Then $P(A) = 0.3$, $P(B) = 0.4$, $P(A') = 0.7$, $P(B') = 0.6$. Also $P(A \cap B) = 0.1$, $P(A \cup B) = 0.6$. Clearly:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \Rightarrow 0.6 = 0.3 + 0.4 - 0.1$$

A Venn diagram may also contain probabilities instead of numbers of elements. The Venn diagram above takes the form:



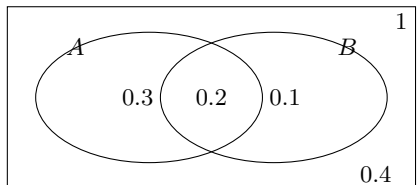
Mutually exclusive sets (probability)



Two events are mutually exclusive if $A \cap B = \emptyset$, equivalently $n(A \cap B) = 0$, equivalently $P(A \cap B) = 0$. In this case only:

$$P(A \cup B) = P(A) + P(B)$$

Example 1. Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \cup B) = 0.6$. Construct a Venn diagram. Notice $P(A \cup B) \neq P(A) + P(B)$ since $0.6 \neq 0.8$. The difference implies an intersection exists: $P(A \cap B) = 0.2$. Starting from the intersection 0.2 , we may easily complete the following Venn diagram:



After completing the Venn diagram, we are in a position to answer any probability question. For example:

$$\begin{aligned} P(A \cap B') &= 0.3 & P(A' \cap B) &= 0.1 & P(A' \cap B') &= 0.4 \\ P(A \cup B') &= 0.9 & P(A' \cup B) &= 0.7 & P(A' \cup B') &= 0.8 \end{aligned}$$

Tables

Another way to represent sets in order to find probabilities — appropriate when the sample space is partitioned by two different criteria (e.g. MALE-FEMALE and SMOKER-NON SMOKER).

Example. a group of 200 people:

	male	female	Total
smoker	40	20	60
non-smoker	80	60	140
Total	120	80	200

Selecting a person at random:

$$\begin{aligned}
P(\text{male}) &= 120/200 = 0.6 & P(\text{female}) &= 80/200 = 0.4 \\
P(\text{smoker}) &= 60/200 = 0.3 & P(\text{non-smoker}) &= 140/200 = 0.7 \\
P(\text{male AND smoker}) &= 40/200 = 0.2 \\
P(\text{male OR smoker}) &= 140/200 = 0.7
\end{aligned}$$

Notice: for “male OR smoker” we consider the male column and the smoker row, but the male-smoker combination is counted only once. It holds again:

$$P(\text{male} \cup \text{smoker}) = P(\text{male}) + P(\text{smoker}) - P(\text{male} \cap \text{smoker})$$

Two dice

Some problems require particular counting techniques — tossing two dice is a classic example: there are **36** possible outcomes (a 6×6 table of combinations). The following diagram helps to visualize the outcomes (each dot represents one outcome; the dashed line marks the sum-of-scores-equal-to-9 diagonal):

	1	2	3	4	5	6
1	•	•	•	•	•	•
2	•	•	•	•	•	•
3	•	•	•	•	•	•
4	•	•	•	•	•	•
5	•	•	•	•	•	•
6	•	•	•	•	•	•

There is only one combination of “double one” (1-1), but two combinations of “one-two” (1-2 and 2-1). Typical probabilities found this way:

$$\begin{aligned}
P(\text{two sixes}) &= 1/36 && \text{(the very last cell)} \\
P(\text{at least one six}) &= 11/36 && \text{(last row and last column)} \\
P(\text{exactly one six}) &= 10/36 \\
P(\text{same score}) &= 6/36 && \text{(the main diagonal)} \\
P(\text{sum of scores} < 9) &= 26/36 \\
P(\text{sum of scores} = 9) &= 4/36 \\
P(\text{sum of scores} > 9) &= 6/36
\end{aligned}$$

3 4.7 Conditional Probability – Independent Events

Notation:

$P(A)$ means ‘probability of A’

$P(A|B)$ means ‘probability of A, given B’

Example. pick a whole number in the range 1–100.

$A = \text{‘‘we pick 17’’}$ $P(A) = 1/100$

$B = \text{‘‘the number selected has two digits’’}$

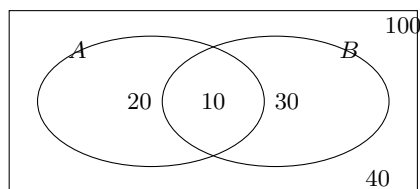
$P(A|B) = 1/90$ (there are 90 two-digit numbers)

Formal definition of $P(A|B)$

$$P(A|B) = n(A \cap B) / n(B) \quad \text{or equivalently} \quad P(A|B) = P(A \cap B) / P(B)$$

$P(A|B)$ in a Venn diagram

Example. consider:



We know $P(A) = 30/100$. What about $P(A|B)$? Given B , the new ‘total’ is 40 (the size of B), not 100. How many elements of A lie in B ? Only 10, so:

$$P(A|B) = 10/40 = 0.25$$

Check via the formula: $P(A|B) = n(A \cap B)/n(B) = 10/40 = 0.25$. Dividing numerator and denominator by TOTAL gives the formal definition: $P(A|B) = (10/100)/(40/100) = 0.25$.

Similarly:

$$P(B|A) = 10/30 \quad (\text{given } A)$$

$$P(A'|B) = 30/40 \quad P(A|B') = 20/60 \quad P(A'|B') = 40/60$$

$P(A|B)$ in a table

Using the smoker table above:

	male	female	Total
smoker	40	20	60
non-smoker	80	60	140
Total	120	80	200

$$P(\text{smoker}) = 60/200$$

$$P(\text{smoker}|\text{male}) = 40/120 \quad (\text{given male})$$

$$P(\text{male}|\text{smoker}) = 40/60 \quad (\text{given smoker})$$

$$P(\text{female}|\text{smoker}) = 20/60 \approx 0.33$$

$$P(\text{non-smoker}|\text{female}) = 60/80 = 0.75$$

Independent events

Events A and B are **independent** if:

$$P(A|B) = P(A)$$

i.e. B does not affect A — the probability of A stays the same whether B is given or not. Similarly, in this case $P(B|A) = P(B)$, i.e. A does not affect B .

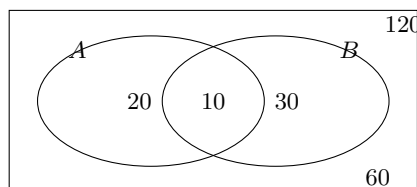
Since $P(A|B) = P(A \cap B)/P(B)$, independence gives:

$$P(A \cap B) = P(A|B) \cdot P(B) = P(A) \cdot P(B)$$

To summarize, A and B are independent if any one (equivalently all) of these hold:

- (1) $P(A|B) = P(A)$
- (2) $P(B|A) = P(B)$
- (3) $P(A \cap B) = P(A) \cdot P(B)$

Example 1. Consider:



We can show in three different ways that A and B are independent:

$$\begin{array}{llll}
 P(A) = 30/120 = 1/4 & \text{and} & P(A|B) = 10/40 = 1/4 & \Rightarrow (1) \text{ holds} \\
 P(B) = 40/120 = 1/3 & \text{and} & P(B|A) = 10/30 = 1/3 & \Rightarrow (2) \text{ holds} \\
 P(A \cap B) = 10/120 = 1/12 & \text{and} & P(A) \cdot P(B) = (1/4)(1/3) = 1/12 & \Rightarrow (3) \text{ holds}
 \end{array}$$

Notice — a common confusion

Many students confuse “mutually exclusive” with “independent”:

Mutually exclusive events means $A \cap B = \emptyset$

Independent events means $P(A \cap B) = P(A) \cdot P(B)$

Also:

$$\begin{array}{ll}
 P(A \cup B) = P(A) + P(B) - P(A \cap B) & \text{holds in general} \\
 P(A \cap B) = P(A) \cdot P(B) & \text{holds for independent events}
 \end{array}$$

Combining them for independent events gives:

$$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$$

Sometimes we know beforehand that two events are independent, so we apply $P(A \cap B) = P(A) \cdot P(B)$ directly.

Example. toss a die and a coin. Find the probability the die shows a SIX and the coin shows a HEAD.

$$\begin{aligned} A &= \text{“die shows SIX”} & P(A) &= 1/6 \\ B &= \text{“coin shows HEAD”} & P(B) &= 1/2 \end{aligned}$$

A and B are clearly independent, so: $P(A \cap B) = P(A) \cdot P(B) = (1/6)(1/2) = 1/12$.

Example 2. Let $P(A) = 0.4$ and $P(B) = 0.3$. Find $P(A \cup B)$ in each case:

- (a) A and B are mutually exclusive: $P(A \cup B) = P(A) + P(B) = 0.4 + 0.3 = 0.7$
- (b) A and B are independent: $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B) = 0.4 + 0.3 - 0.12 = 0.58$
- (c) $P(A \cap B) = 0.2$: $P(A \cup B) = 0.4 + 0.3 - 0.2 = 0.5$
- (d) $P(A|B) = 0.2$: $P(A \cap B) = P(A|B) \cdot P(B) = (0.2)(0.3) = 0.06$, so $P(A \cup B) = 0.4 + 0.3 - 0.06 = 0.64$

Example 3. A and B are independent, with $P(A) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(B)$.

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A) \cdot P(B) \\ 0.7 &= 0.4 + P(B) - 0.4 \cdot P(B) \\ 0.3 &= 0.6 \cdot P(B) \\ P(B) &= 0.5 \end{aligned}$$

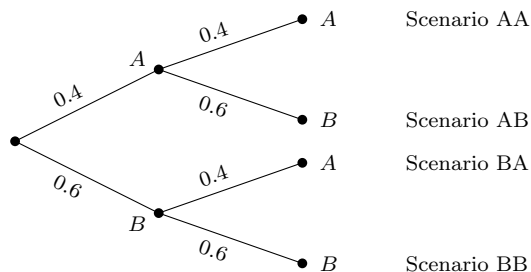
4 4.8 Tree Diagrams

A tree diagram is the best way to represent a sequence of events under different scenarios. In each scenario, we multiply the probabilities along the corresponding path; the sum of all final (leaf) probabilities of a tree is always **1**.

Problem 1. We play a game with two possible results: pick one of the letters from AAAA BBBB (10 letters), so:

A with probability 0.4 B with probability 0.6

Playing the game **twice**, all possible scenarios are shown below; the corresponding probabilities are shown on the branches of the tree:



Next, for each scenario we multiply the corresponding probabilities:

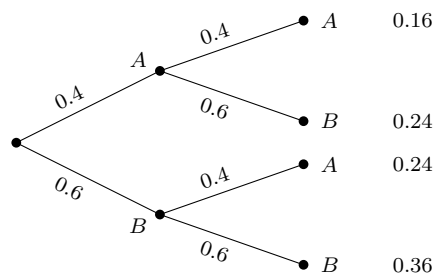
$$P(AA) = 0.4 \times 0.4 = 0.16$$

$$P(AB) = 0.4 \times 0.6 = 0.24$$

$$P(BA) = 0.6 \times 0.4 = 0.24$$

$$P(BB) = 0.6 \times 0.6 = 0.36$$

Thus, the final “picture” of the tree diagram is as follows:



(notice that the sum of the resulting probabilities is 1). We can then answer any probability question by adding the relevant leaves:

$$P(\text{two A's}) = 0.16 \quad P(\text{two B's}) = 0.36$$

$$P(\text{first A, then B}) = 0.24$$

$$P(\text{one A, one B}) = 0.24 + 0.24 = 0.48$$

$$P(\text{same result twice}) = 0.16 + 0.36 = 0.52 \quad (\text{so different results: } 1 - 0.52 = 0.48)$$

In terms of the number of A's obtained:

$$P(\text{no A}) = 0.36$$

$$P(\text{exactly one A}) = 0.24 + 0.24 = 0.48$$

$$P(\text{at least one A}) = 0.24 + 0.24 + 0.16 = 0.64$$

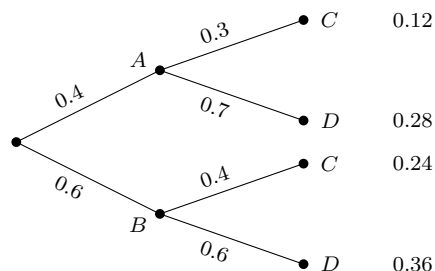
$$P(\text{at most one A}) = 0.24 + 0.24 + 0.36 = 0.84$$

Problem 2. We play the previous game once, then, depending on the result, play a *different* second game:

If the first result is A, pick a letter among CCC DDDDDDD ($P(C|A)=0.3$, $P(D|A)=0.7$)

If the first result is B, pick a letter among CCCC DDDDDD ($P(C|B)=0.4$, $P(D|B)=0.6$)

A tree diagram is particularly helpful in such a situation where the second game depends on the first one:



(notice again that the sum of the resulting probabilities is 1). So:

$$P(C) = 0.12 + 0.24 = 0.36 \quad P(D) = 0.28 + 0.36 = 0.64$$

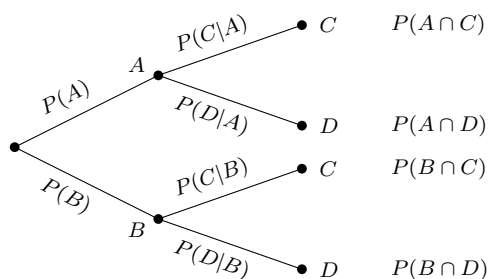
Also worth noting:

$$P(A \text{ and } C) = P(A \cap C) = 0.12 \quad P(A \text{ or } C) = P(A \cup C) = 0.12 + 0.28 + 0.24 = 0.64$$

Notice — structure of a tree diagram

In general:

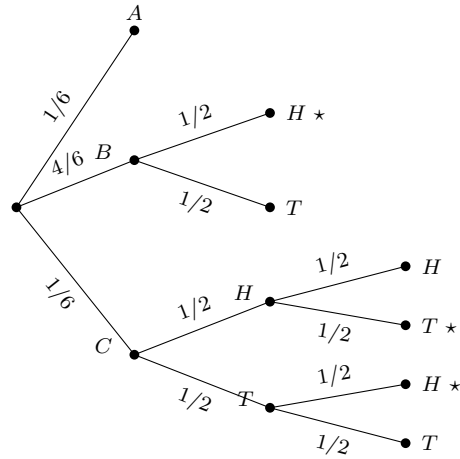
- 1st-column branches carry simple probabilities $P(X)$
- 2nd-column branches carry conditional probabilities $P(Y|X)$
- the leaves carry combined probabilities $P(X \cap Y)$



Trees may be more complex, with more branches per level and more levels.

Example 1. We throw a die. If we get 1 we stop. If we get 2,3,4 or 5 we toss a coin. If we get 6 we toss two coins. Find the probability that only one head is obtained.

Solution. For our convenience, we denote the results of the die by $A = \{1\}$, $B = \{2, 3, 4, 5\}$, $C = \{6\}$. We construct the following tree diagram (branches marked $\star$ give exactly one HEAD):

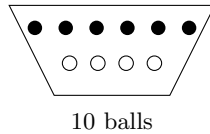


There are finally 7 scenarios (seven paths). In 3 of them we have exactly one HEAD (marked $\star$). Adding the corresponding results:

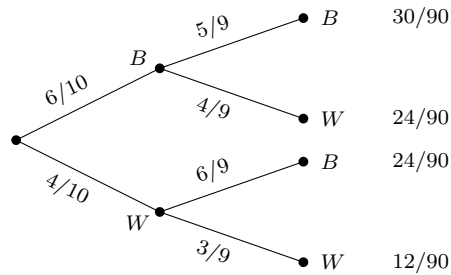
$$P(\text{only one HEAD}) = \frac{4}{6} \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{5}{12}$$

A typical example: colored balls in a box (sampling without replacement)

A box contains 10 balls: 6 BLACK and 4 WHITE:



We select two balls, one after the other. All possible outcomes are clearly shown on the following tree diagram:



$$P(\text{both balls BLACK}) = 30/90 = 1/3$$

$$P(\text{only one ball BLACK}) = 24/90 + 24/90 = 48/90 = 8/15$$

$$P(\text{balls of same color}) = 30/90 + 12/90 = 42/90 = 7/15$$

For **3 balls**, the same reasoning works directly without drawing the full tree:

$$P(\text{all three BLACK}) = (6/10)(5/9)(4/8) = 1/6$$

$$P(\text{only one ball BLACK}) = 3 \times (6/10)(4/9)(3/8) = 3/10$$

The “reverse given”

Using the tree of Problem 2 above, we know $P(C|A) = 0.3$ directly from the tree. What about $P(A|C)$ — given that the *final* result is C , what is the probability the *first* result was A ? This is not shown directly on the tree; it is:

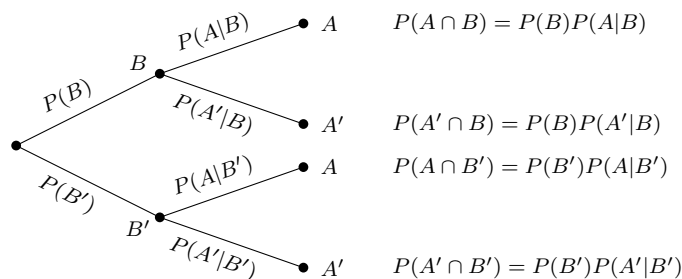
$$P(A|C) = 0.12/(0.12 + 0.24) = 0.12/0.36 \approx 0.33$$

which is exactly the formula $P(A|C) = P(A \cap C)/P(C)$. Similarly:

$$P(B|C) = 0.24/0.36 \approx 0.67 \quad P(A|D) = 0.28/0.64 \approx 0.44 \quad P(B|D) = 0.36/0.64 \approx 0.56$$

Bayes’ Theorem (HL only)

The “reverse given” result is formally known as **Bayes’ Theorem**, and its formula can be obtained directly from a tree diagram. The events in the first column are called B and B' ; the events in the second column are called A and A' :



$$P(B|A) = \frac{P(B) \cdot P(A|B)}{P(B) \cdot P(A|B) + P(B') \cdot P(A|B')}$$

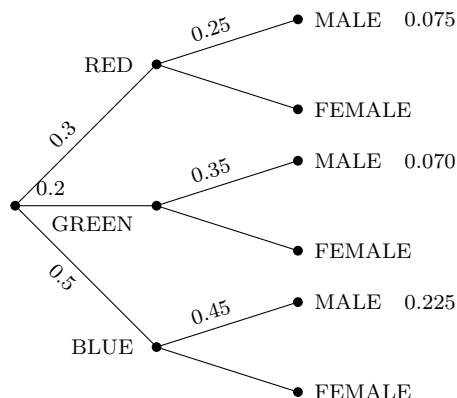
If the first level of the tree has 3 disjoint events B_1, B_2, B_3 , then:

$$P(B_1|A) = \frac{P(B_1) \cdot P(A|B_1)}{P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3)}$$

(similar formulas for $P(B_2|A)$ and $P(B_3|A)$).

Example 2. At a private school party, 30% of students wear RED, 20% wear GREEN, 50% wear BLUE. Of the RED students 25% are MALE, of the GREEN 35% are MALE, of the BLUE 45% are MALE. Find $P(\text{GREEN}|\text{MALE})$.

Instead of the Bayes formula directly, construct a tree (notice that we do not complete all the probabilities on the tree diagram, but only the “necessary” ones):



$$\begin{aligned}\text{RED} \cap \text{MALE} &= 0.30 \times 0.25 = 0.075 \\ \text{GREEN} \cap \text{MALE} &= 0.20 \times 0.35 = 0.070 \\ \text{BLUE} \cap \text{MALE} &= 0.50 \times 0.45 = 0.225\end{aligned}$$

Then:

$$P(\text{GREEN}|\text{MALE}) = 0.070 / (0.075 + 0.070 + 0.225) = 0.070 / 0.37 \approx 0.189$$

i.e. 18.9% of MALE students wear GREEN suits.

5 4.9 Discrete Distributions in General

A random variable X takes values at random in a given domain — it may be **discrete** (e.g. $X \in \{10, 20, 30\}$ or $X \in \{0, 1, 2, 3, \dots\}$) or **continuous** (e.g. $X \in [10, 20]$ or $X \in \mathbb{R}$). A discrete variable takes values in a finite or numerable set; a continuous variable takes values in some interval(s). This section deals only with discrete random variables.

Discrete random variable

Example. X takes values 10, 20, 30 with probabilities 0.2, 0.3, 0.5:

x	10	20	30
$P(X = x)$	0.2	0.3	0.5

Clearly (i) all probabilities are non-negative, and (ii) their sum is always 1. Then X is a **discrete random variable**. We write $P(X=10) = 0.2$, $P(X=20) = 0.3$, $P(X=30) = 0.5$.

In general, for a discrete random variable with values $x_1, x_2, x_3, \dots$ and probabilities $p_1, p_2, p_3, \dots$:

- (i) $p_i \geq 0$ for all i
- (ii) $\sum p_i = 1$ i.e. $p_1 + p_2 + p_3 + \dots = 1$

(We also say a probability function $p : x_i \mapsto p_i$ is defined.)

The expected value $\mu = E(X)$

$$E(X) = \sum x_i p_i = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots$$

For the table above: $E(X) = 10(0.2) + 20(0.3) + 30(0.5) = 23$.

Notice — explanation for $\mu = E(X)$. This mean is not different from the mean in statistics. Consider the ten numbers 10, 10, 20, 20, 20, 30, 30, 30, 30, 30 (with the same relative frequencies as the probability table above). The statistical mean is:

$$\mu = \frac{10 \times 2 + 20 \times 3 + 30 \times 5}{10} = 10 \left(\frac{2}{10} \right) + 20 \left(\frac{3}{10} \right) + 30 \left(\frac{5}{10} \right) = 23$$

Example 1. Given the table x : 10, 20, 30 with $P(X=x)$: a , b , 0.5, and $E(X) = 23$, find a and b .

$$\begin{aligned} a + b + 0.5 &= 1 & \Rightarrow & a + b = 0.5 \\ 10a + 20b + 15 &= 23 & \Rightarrow & 10a + 20b = 8 \end{aligned}$$

Solving: $a = 0.2$, $b = 0.3$.

The probability distribution applies to many betting games.

Example 2. Using the same table ($x = 10, 20, 30$; $P = 0.2, 0.3, 0.5$): selecting 10 earns 6 points, 20 earns 1 point, 30 loses 2 points. What is the expected profit in one game?

x	10	20	30
Profit	6 pts	1 pt	-2 pts
Prob	0.2	0.3	0.5

$$\text{Expected profit} = 6(0.2) + 1(0.3) - 2(0.5) = 0.5$$

i.e. on average we earn 0.5 points per game. (Explanation: over 10 games we'd expect $2 \times 6 + 3 \times 1 - 5 \times 2 = 12 + 3 - 10 = 5$ points, i.e. 0.5/game.)

Example 3. Throw two dice. TWO SIXES earns €15, ONE SIX earns €1, NO SIX loses €1. Find the expected profit.

Result	TWO SIXES	ONE SIX	NO SIX
Profit	15€	1€	-1€
Prob	1/36	10/36	25/36

$$\text{Expected profit} = 15 \left(\frac{1}{36} \right) + 1 \left(\frac{10}{36} \right) - 1 \left(\frac{25}{36} \right) = 0$$

This is a **FAIR GAME** — we expect to neither win nor lose. (Note: if the top prize were €14 instead of €15, the expected profit would be $-1/36$, i.e. over 36,000 games/€ we'd expect to lose €1,000.)

Median – Mode

Defined analogously to statistics:

MODE = the value $X=a$ with the highest probability

MEDIAN = the value $X=m$ where the probability splits into two equal halves (0.5--0.5)

Examples. Look at the tables below:

x	10	20	30
$P(X = x)$	0.4	0.3	0.3

MODE = 10

MEDIAN = 20

x	10	20	30
$P(X = x)$	0.2	0.3	0.5

MODE = 30

MEDIAN = 25 (why?)

Variance (HL only)

$$\text{Var}(X) = E[(X - \mu)^2] = (x_1 - \mu)^2 p_1 + (x_2 - \mu)^2 p_2 + (x_3 - \mu)^2 p_3 + \dots$$

Equivalent definition:

$$\text{Var}(X) = E(X^2) - \mu^2 \quad \text{where} \quad E(X^2) = x_1^2 p_1 + x_2^2 p_2 + x_3^2 p_3 + \dots$$

Example 4. For x : 10,20,30, P : 0.2,0.3,0.5, we found $\mu = 23$. Then:

$$\text{Var}(X) = (10 - 23)^2(0.2) + (20 - 23)^2(0.3) + (30 - 23)^2(0.5) = 61$$

Or: $E(X^2) = 10^2(0.2) + 20^2(0.3) + 30^2(0.5) = 590$, so $\text{Var}(X) = 590 - 23^2 = 61$.

6 4.10 Binomial Distribution – B(n,p)

It is the distribution of a discrete random variable X taking values $0, 1, 2, \dots, n$, with probability function:

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, 3, \dots, n$$

where n and p are the two parameters. (*This formula is not in the syllabus — results are obtained directly by GDC — but it is worth knowing how it works.*)

Description of the problem

We deal with a game (or experiment) with two outcomes: SUCCESS with probability p , FAILURE with probability $1 - p$. We play n times:

n = number of trials **p** = probability of success
X = number of (possible) successes

X follows a binomial distribution: $\mathbf{X} \sim \mathbf{B(n, p)}$, taking values $0, 1, 2, \dots, n$.

GDC

MENU -- Statistics -- DIST -- BINOMIAL: use Bpd or Bcd
 Bpd(x) = probability of exactly x successes
 Bcd(x1 to x2) = probability from x1 up to x2 successes
 Menu fields: Data (always Variable), Numtrial (=n), p

Example 1. Toss a die 5 times; success = getting a six, so $n = 5$, $p = 1/6$. X can be 0 to 5:

x	0	1	2	3	4	5
$P(X = x)$	0.4019	0.4019	0.1608	0.0322	0.0032	0.0001

Further questions:

Find the probability of	Notation	GDC	Result
exactly 3 sixes	$P(X = 3)$	Bpd(3)	0.0322
at most 3 sixes	$P(X \leq 3)$	Bcd(0 to 3)	0.9967
less than 3 sixes	$P(X < 3)$	Bcd(0 to 2)	0.9645
more than 3 sixes	$P(X > 3)$	Bcd(4 to 5)	0.0033
at least 3 sixes	$P(X \geq 3)$	Bcd(3 to 5)	0.0355

Remark on the formula (not in syllabus, worth knowing). The probability of:

5 sixes in a row: $1/6^5$
 no six at all: $5^5/6^5$
 2 sixes and 3 no-sixes: $\binom{5}{2}(1/6)^2(5/6)^3$

where $\binom{5}{2}$ (also written 5C2) is the number of ways to choose 2 sixes among 5 trials. In general, the probability of x sixes (and $5 - x$ no-sixes) is $\binom{5}{x}(1/6)^x(5/6)^{5-x}$, and more generally for n trials

with success probability p :

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Applying the formula for the die example reproduces the same table as Bpd(x) above (e.g. $P(X=0) = 3125/7776 \approx 0.4019$, etc.).

Expected value and variance of X

$$E(X) = np \quad \text{Var}(X) = np(1-p)$$

For the die example: $E(X) = 5(1/6) = 5/6$, $\text{Var}(X) = 5(1/6)(5/6) = 25/36$.

Notice (HL only). Since $E(X)$ and $\text{Var}(X)$ are known, $E(X^2)$ follows:

$$\text{Var}(X) = E(X^2) - E(X)^2 \implies E(X^2) = \text{Var}(X) + E(X)^2 = np(1-p) + (np)^2$$

Example 2. A box has 5 balls: 1 BLACK, 4 WHITE. We win if we select the BLACK ball, and play 10 times. X = number of wins, so $X \sim B(10, 0.2)$ ($p = 1/5 = 0.2$). Find:

- (a) $P(\text{exactly 4 wins}) = \text{Bpd}(4) = 0.088$ [indeed $\binom{10}{4}(0.2)^4(0.8)^6 = 0.088$]
- (b) $P(\text{at most 4 wins}) = \text{Bcd}(0 \text{ to } 4) = 0.967$ [= $P(X=0) + \dots + P(X=4)$]
- (c) $P(\text{at least 1 win}) = \text{Bcd}(1 \text{ to } 10) = 0.893$ [= $1 - P(X=0) = 1 - 0.107$]
- (d) $E(X) = np = 10(0.2) = 2$
- (e) $\text{Var}(X) = np(1-p) = 10(0.2)(0.8) = 1.6$

Example 3. $p = 0.2$, n unknown. Given $P(X=1) = 0.268$, find n . Since n must be an integer, trial and error on Numtrial shows $\text{Bpd}(10)=0.268$ when $n = 10$. Hence $n = 10$.

Mode (mainly for HL)

First check the expected value:

- **If $E(X) = np$ is not a whole number** (e.g. $n = 20$, $p = 1/6$, $E(X) = 20/6 \approx 3.3$): check the two nearest integers, e.g. $P(X=3) = 0.237$, $P(X=4) = 0.202 \rightarrow \text{mode} = 3$ (highest probability).
- **If $E(X)$ is a whole number** (e.g. $n = 60$, $p = 1/6$, $E(X) = 60/6 = 10$): check the neighboring integers 9,10,11, e.g. $P(X=9) = 0.134$, $P(X=10) = 0.137$, $P(X=11) = 0.126 \rightarrow \text{mode} = 10$.

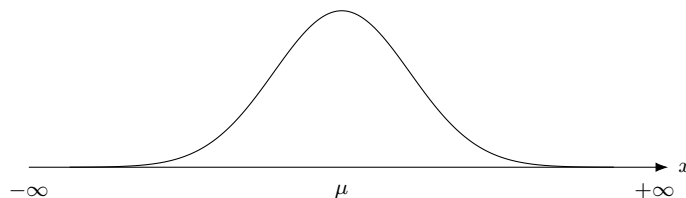
Notice — sometimes there are two modes. For $n = 5$, $p = 1/6$, $E(X) = 5/6 \approx 0.833$: check $P(X=0) = 0.4019$ and $P(X=1) = 0.4019$ — equal, so there are **two modes**, 0 and 1.

7 4.11 Normal Distribution – $N(\mu, \sigma^2)$

It is the distribution of a **continuous** random variable X with values from $-\infty$ to $+\infty$. Parameters:

μ = mean σ = standard deviation

The probability “behaves” as a symmetric bell-shaped curve peaking at μ :



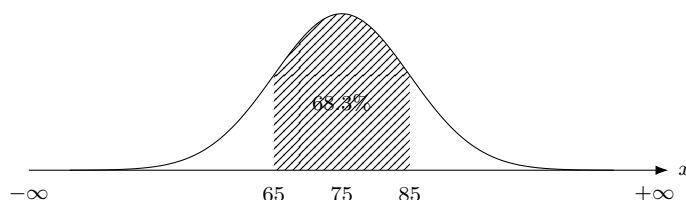
There is a highly likely mean value μ , and values spread out symmetrically about the mean; as we move away from μ (left or right) the probability decreases dramatically. We write $\mathbf{X} \sim \mathbf{N}(\mu, \sigma^2)$.

Description of the problem in general

It is the most “popular” distribution in nature. Random variables depending on many factors follow it, e.g.: weight of people, height of people, time spent in a supermarket, weight of a labeled coffee pack.

Example. for a Greek man, mean weight $\mu = 75\text{kg}$, st.dev. $\sigma = 10\text{kg}$. It is estimated that:

Percentage of population	Range (general)	Range (this example)
$\approx 68\%$	$\mu - \sigma$ to $\mu + \sigma$	[65, 85]
$\approx 95\%$	$\mu - 2\sigma$ to $\mu + 2\sigma$	[55, 95]
$\approx 99.7\%$	$\mu - 3\sigma$ to $\mu + 3\sigma$	[45, 105]



Notice:

- The whole area under the curve is 1 (100%). The area before the mean and after the mean are each 0.5 (50%).
- Theoretically X ranges from $-\infty$ to $+\infty$; in practice, almost the whole population (99.7%) ranges between $\mu - 3\sigma$ and $\mu + 3\sigma$.
- σ indicates the spread: e.g. if Greeks have $\mu = 75\text{kg}$, $\sigma = 10\text{kg}$, and Italians have $\mu = 75\text{kg}$, $\sigma = 8\text{kg}$, both populations share the same mean, but Italians are more concentrated around it ($\mu \pm 3\sigma$ gives 45–105kg for Greeks vs. 51–99kg for Italians).

We distinguish three types of problems, all solved via GDC:

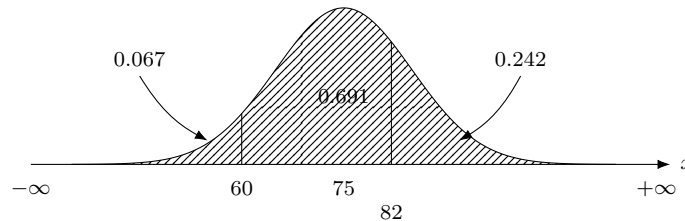
MENU -- STAT -- DIST -- NORM: use Ncd or InvN (Data: always Variable)
 Ncd is used when we ask for a probability
 InvN is used when we know the probability

Problem 1: Find probability (μ, σ known — use Ncd)

Example. X = weight of a Greek man, $\mu = 75\text{kg}$, $\sigma = 10\text{kg}$. Find:

- (a) $P(60 \leq X \leq 82) = 0.691$
- (b) $P(X \geq 82) = 0.242$
- (c) $P(X \leq 60) = 0.067$

Let us represent the information of this problem by a diagram:



Notice:

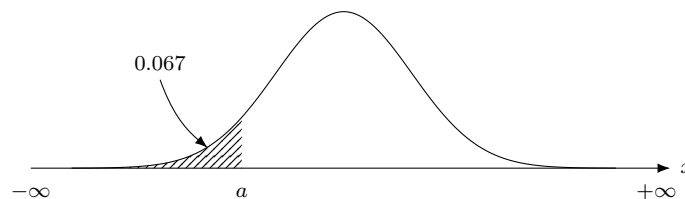
- For $P(60 \leq X \leq 82) = 0.691$, the GDC also shows $z:\text{Low} = -1.5$, $z:\text{Up} = 0.7$, meaning: the lower bound is 1.5 st.devs below μ ($75 - 1.5 \times 10 = 60$), and the upper bound is 0.7 st.devs above μ ($75 + 0.7 \times 10 = 82$). These are called **standardized values** (see below).
- The probability that the weight is 1 st.dev away from the mean (between 65 and 85kg) is $P(65 \leq X \leq 85) \approx 0.683$ (68.3%), with $z:\text{Low} = -1$, $z:\text{Up} = 1$.
- These probabilities refer to selecting **one** person. For two people:

$$\begin{aligned} P(\text{both between 60 and 82kg}) &= (0.691)^2 \\ P(\text{neither between 60 and 82kg}) &= (0.309)^2 \\ P(\text{only one between 60 and 82kg}) &= 2(0.691)(0.309) \end{aligned}$$

- A question may combine normal and binomial distributions: if we select 10 people, the probability that exactly 3 are between 60 and 82kg uses the normal result $p = 0.691$ as the success probability of a new binomial variable $Y \sim B(10, 0.691)$: $P(Y=3) = 0.0106$.

Problem 2: Probability is given (μ, σ known — use InvN)

Again $\mu = 75\text{kg}$, $\sigma = 10\text{kg}$ for X = weight of a Greek man. Given $P(X \leq a) = 0.067$, find a . (Equivalently: “6.7% of Greek men weigh less than a . Find a .”) Let us represent this information in a diagram:



Using InvN with Tail: Left, Area: 0.067, $\sigma = 10$, $\mu = 75 \rightarrow a = 60\text{kg}$.

Notice for the tail:

Tail: Left $\rightarrow$ if we know the area BEFORE some value

Tail: Right $\rightarrow$ if we know the area AFTER some value

The same answer can be obtained via right tail: the area after a is $1 - 0.067 = 0.933$, i.e. $P(X \geq a) = 0.933$, so Tail: Right, Area: 0.933 also gives $a = 60\text{kg}$.

Example 1. The mass of coffee packs is normally distributed with mean 500g, st.dev. 15g.

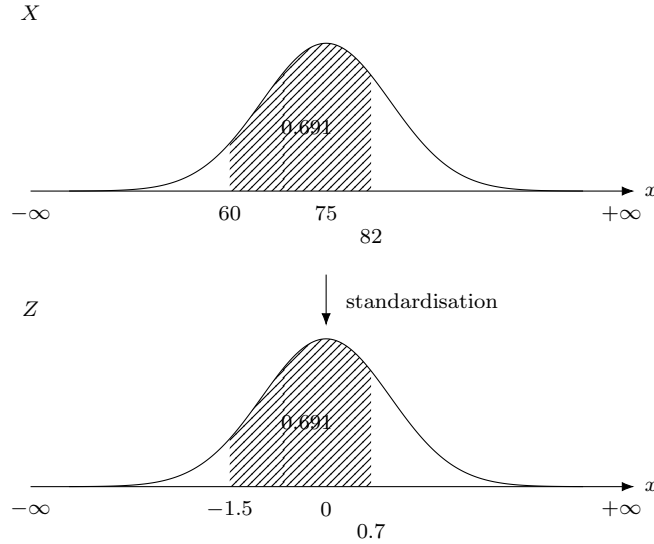
- (a) $P(X > 520) \approx 0.091$ (via Ncd)
- (b) The lightest 4% weigh less than a , the heaviest 4% weigh more than b (via InvN): $P(X \leq a) = 0.04 \rightarrow a \approx 474\text{g}$; $P(X \geq b) = 0.04 \rightarrow b \approx 526\text{g}$.
- (c) Packs in (b) are rejected. In a daily production of 1600 packs, 8% is rejected $\rightarrow 1600 \times 0.08 = 128$ packs expected rejected.
- (d) $P(\text{both of 2 selected packs rejected}) = (0.08)^2 = 0.0064$.
- (e) For 5 selected packs, $P(\text{at least one rejected})$: binomial with $n = 5$, $p = 0.08 \rightarrow P(X \geq 1) \approx 0.341$.
- (f) Find Q_1 , Q_3 (lower/upper quartiles): area before Q_1 is 0.25, before Q_3 is 0.75, via InvN $\rightarrow Q_1 \approx 490\text{g}$, $Q_3 \approx 510\text{g}$ (for this symmetric case, Tail: Central, Area=0.5 also works).

Standardisation – Normal distribution $N(0,1)$

The random variable Z following normal distribution with $\mu = 0$, $\sigma = 1$ is the **standardised normal distribution**: $Z \sim N(0,1)$. Any $X \sim N(\mu, \sigma^2)$ can be transformed into Z via:

$$Z = \frac{X - \mu}{\sigma}$$

Example. $X \sim N(75, 10^2)$. Then $Z = (X - 75)/10 \sim N(0,1)$. Standardized value of $x = 60$: $z_{60} = (60 - 75)/10 = -1.5$. Standardized value of $x = 82$: $z_{82} = (82 - 75)/10 = 0.7$. Check: $P(60 \leq X \leq 82) = 0.691$ and $P(-1.5 \leq Z \leq 0.7) = 0.691$. In diagrams:



Problem 3: Find μ or σ (use Standardisation and InvN)

$X \sim N(150, \sigma^2)$ with σ unknown; given $P(X \leq 140) = 0.25$ (i.e. 25% is less than 140), find σ .

Using $Z = (X - \mu)/\sigma$, first find Z_{140} via InvN with Tail: Left, Area: 0.25, $\sigma = 1$, $\mu = 0 \rightarrow Z_{140} = -0.6745$. Then:

$$-0.6745 = \frac{140 - 150}{\sigma} = \frac{-10}{\sigma} \Rightarrow \sigma = \frac{-10}{-0.6745} \approx 14.83$$

A similar process applies if instead μ is unknown: for $\sigma = 14.83$, μ unknown, the standardized value Z_{140} is the same (-0.6745):

$$-0.6745 = \frac{140 - \mu}{14.83} \Rightarrow 140 - \mu = -10 \Rightarrow \mu = 150$$

Example 2. For X , it's known that 35% is less than 60 and 25% is more than 90, i.e. $P(X \leq 60) = 0.35$, $P(X \geq 90) = 0.25$. Find μ and σ .

Using InvN: Tail:Left, Area:0.35, $\sigma = 1$, $\mu = 0 \rightarrow Z_{60} = -0.385$. Tail:Right, Area:0.25, $\sigma = 1$, $\mu = 0 \rightarrow Z_{90} = 0.674$. This gives the system:

$$\begin{aligned} -0.385 &= \frac{60 - \mu}{\sigma} & \Rightarrow \mu - 0.385\sigma &= 60 \\ 0.674 &= \frac{90 - \mu}{\sigma} & \Rightarrow \mu + 0.674\sigma &= 90 \end{aligned}$$

Solving: $\mu \approx 70.9$, $\sigma \approx 28.3$.

Notice (HL only). Since you know $E(X)$ and $\text{Var}(X)$, you also know $E(X^2)$. Indeed:

$$\text{Var}(X) = E(X^2) - E(X)^2 \Rightarrow E(X^2) = \text{Var}(X) + E(X)^2 = \sigma^2 + \mu^2$$

End of theory and examples for sections 4.5–4.11.